

Jonathan Mehmannaavaz, CEP

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EDUCATION

Bachelor of Engineering in Computer Engineering

Sep 2022 – Apr 2026

Concordia University

Montreal, QC

- Relevant Coursework: Artificial Intelligence, Applied Machine Learning & Evolutionary Algorithms, Embedded Systems, Cloud Computing, Communication Networks & Protocols, Digital System Design, Computer Graphics

Technological Academic Path in Engineering

2021 – 2022

École de technologie supérieure (ETS)

Montreal, QC

- Relevant Coursework: Programmable Logic Controllers (PLCs), Sequential Logic, Computer & Technical Drawing for Engineers

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, Swift, HTML/CSS, PL-SQL, VHDL, Bash, Assembly

Frameworks & Libraries: Flask, React, Node.js, Express.js, Streamlit, gRPC, Protocol Buffers, TensorFlow, Keras, Scikit-learn

Cloud/DevOps: Docker, Azure, Google Cloud, MongoDB, PostgreSQL, Redis, RabbitMQ, GitHub Actions

Tools & Hardware: Git, Linux, Arduino, FPGA, ARM Cortex-M, PLC, MATLAB, SolidWorks, Altium Designer, ModelSim, Power BI, SAP

Human Languages: French (Native), English (Fluent), German (Beginner)

Professional Designation: Candidate to the Engineering Profession (CEP), Ordre des ingénieurs du Québec (2026)

Certifications: TensorFlow for Deep Learning Bootcamp (Udemy), Power BI (Udemy), iOS Development (Udemy)

EXPERIENCE

Hardware Integration and Data Architect

Jul 2026 – Present

IBM

Bromont, QC

- Design and implement the plant's OT-IT data architecture, integrating equipment data into analytical systems
- Develop edge solutions with Node-RED for industrial data acquisition and communication (MQTT Sparkplug, TCP/IP, APIs)
- Define requirements and integrate the SECS/GEM protocol, from technology selection to production deployment
- Design the manufacturing KPI (OEE) architecture: data ingestion, transformation, modeling, and exposure
- Build data pipelines for advanced analytics and AI/ML use cases, collaborating with IT, automation, and data teams to align with Industry 4.0 principles

Capstone Project Engineer & Project Manager

Sep 2025 – Apr 2026

Fuse Energy | Concordia University (ELEC/COEN 490)

Montreal, QC

- Managed a 6-person team to build a modular industrial control system, coordinating firmware, hardware, and software development
- Wrote real-time embedded firmware with industrial protocols, hardware drivers, and safety control logic on an RTOS

- Built a web app (FastAPI + React) to monitor and control the system in real time

Co-op Supervisor

Sep 2023 – Dec 2023

Pratt & Whitney Canada

St-Hubert, QC

- Proposed and helped set up an RF scanning and KPI tracking system to improve inventory visibility
- Led process improvements and employee training that reduced manufacturing defects
- Supervised 12 unionized employees in aerospace manufacturing, handling daily operations and scheduling

PROJECTS

Event-Driven E-Commerce Microservices | *Python, Flask, MongoDB, RabbitMQ, Docker, Azure*

- Built an event-driven platform with 5 Docker services synced across 2 MongoDB databases using RabbitMQ messaging
- Added traffic splitting (0–100%) to migrate from V1 to V2 with no downtime
- Set up a CI/CD pipeline with GitHub Actions to automatically deploy to Azure

Distributed Real-Time Auction System | *Java, UDP/TCP Sockets, Multithreading, Concurrency*

- Built a server running live auctions over UDP (registration, bidding) and TCP (payments, notifications) with a 20-thread pool for concurrent users
- Made bidding thread-safe using locks and concurrent data structures, with real-time notifications to all connected users

TensorFlow Deep Learning Projects | *Python, TensorFlow, Keras, Scikit-learn, NumPy, Jupyter*

- Trained neural networks on 8+ tasks: regression, classification, computer vision, NLP, and time series forecasting
- Built complete ML pipelines from data preprocessing to model evaluation using confusion matrices and F1 scores

Autonomous Hovercraft | *Arduino, C++, SolidWorks, MATLAB, Embedded Programming*

- Led a 5-person team to design and build an autonomous hovercraft using Arduino, SolidWorks, and MATLAB
- Managed the full project from design to competition, with a successful autonomous navigation demo

Remote-Controlled Tower Crane | *SolidWorks, Altium Designer, Laser Cutting, 3D Printing*

- Worked with a 5-person team to design and build a remote-controlled tower crane, doing PCB design, 3D modelling, and manufacturing
- Took part in all project phases from design to testing, meeting all specifications

Job Finder Web Application | *Python, Streamlit, MongoDB, Google OAuth, REST API*

- Built a job search and tracking app with Google login, two MongoDB databases, and live job listings from the JSearch API
- Added a status workflow (New, Applied, Interview, Offered) with personal search history per user

LEADERSHIP & ACTIVITIES

2nd Place, Civil Volleyball Club – Blockstorm Team, Summer 2025: Runner-up finish in recreational league competition

Regional RSEQ Champion – Football Team (2017-2019): Gold and silver medal winner

International Solidarity Trip – Guatemala (2019): Built infrastructure for underserved communities